Access ALITA

ALITA runs online, protected by Cloudflare. The research team logs in with an e-mail code, peer reviewers with an invitation code. No password, no installation.

RESEARCH TEAM

Dedicated access with your e-mail

At `alita.prodigi-marburg.de`. Only the research team's e-mail addresses are enabled.

PEER REVIEW

Separate access with an invitation code

Reviewers log in at `review.prodigi-marburg.de` with the invitation code from the editor. No name or e-mail needed.

RESEARCH TEAM: LOG IN IN THREE STEPS

1

Open ALITA

Open `alita.prodigi-marburg.de` or the access link you were sent, in any current browser.

2

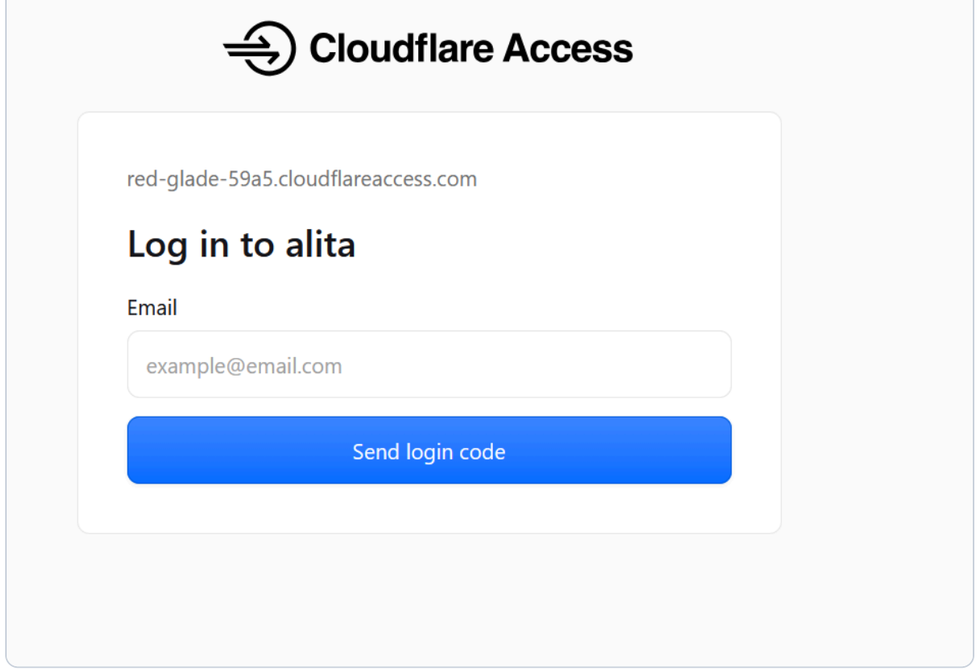
Enter your e-mail

The page **Log in to alita** appears. Enter the address that was enabled for you and click **Send login code**.

3

Enter the code

Cloudflare e-mails you a login code. Enter it on the same page; ALITA opens.



The login page you see in step 2

PEER REVIEWERS: LOG IN WITH THE INVITATION CODE

1

Open the reviewer page

`review.prodigi-marburg.de/review-login.html`

2

Enter the invitation code

Paste the code supplied by the editor into **Invitation code** and click **Continue**.

3

Read and download

Reviewers can read and download all documents but cannot change data.

Keep the invitation code confidential; it is sent separately and not printed here. Cloudflare may process connection information such as your IP address; this is not a guarantee of network anonymity.

WHAT YOU CAN DO ONCE YOU'RE IN

Without an OpenRouter key

Look around:

- Knowledge map, clusters and the 3D view
- Documents list and document details
- Release notes and documentation

With an OpenRouter key with credit

Everything that uses the language model:

- Ask questions in **Document Chat** and **Graph Chat**
- Upload documents with summaries

Remember: each user needs their own key, reviewers too. How to get one → page 2

No login code arriving? (research team)

- Check your spam folder and wait a minute.
- Use exactly the address that was enabled; other addresses get no access.
- Still nothing? Ask the ALITA team to enable your address.

Get an OpenRouter API key

ALITA answers through OpenRouter. You need your own key once; it takes about five minutes.

1 Create an account

Go to `openrouter.ai` and click **Sign in**. Sign up with Google, GitHub or your e-mail address.

2 Add credits

Open `openrouter.ai/settings/credits` and buy credits. OpenRouter works prepaid: requests are paid from this balance.

- Payment by credit card, AliPay or USDC. Card payments carry a 5.5% fee (at least $0.80).
- **Starting with €5 is fine.** At about €0.003 per 10-page paper and well under one cent per question, that lasts for hundreds of questions and uploads.
- Unused credits can be refunded within 24 hours of purchase. Optional: **auto top-up** refills the balance below a threshold you set.

Why credits? ALITA uses a paid model. Free models exist on OpenRouter but are limited to 50 requests per day without purchased credits.

3 Create the key

Open `openrouter.ai/settings/keys` and click **Create Key**.

- **Name:** for example `ALITA`, so you recognise it later.
- **Credit limit:** set a limit (for example $10) or choose no limit. Once the limit is reached, requests with this key are rejected until you raise it.
- **Copy the key right away.** It starts with `sk-or-` and is shown only once.

4 Paste it into ALITA

In the top bar, paste the key into **OpenRouter API-Key** and click **Save**. The status changes from **no key** to **Key …** with its last four characters. The key is stored in this browser; on another device or browser, enter it again.



5 Keep it safe

- Treat the key like a password: don't share it, put it in e-mails or commit it to Git.
- Lost or leaked? Delete it under `openrouter.ai/settings/keys` and create a new one. OpenRouter also e-mails you if it detects an exposed key.
- Check your usage and balance any time under **Activity** and **Credits** in your OpenRouter account.

Checklist

- [ ] Account created
- [ ] Credits added
- [ ] Key copied
- [ ] Saved in ALITA

Fees and limits as stated by OpenRouter, October 2026; they can change. Sources: openrouter.ai/docs/faq · openrouter.ai/docs/api-reference/authentication

Quick Reference

Ask your research documents and get answers with sources you can check.

A · DOCUMENT CHAT
What do the documents say?

B · GRAPH CHAT
How are people, projects and methods connected?

GET STARTED three steps, once

1 Open ALITA

Log in with your e-mail and code (page 1). Pick a view in the top bar: **B** or **A/B**.



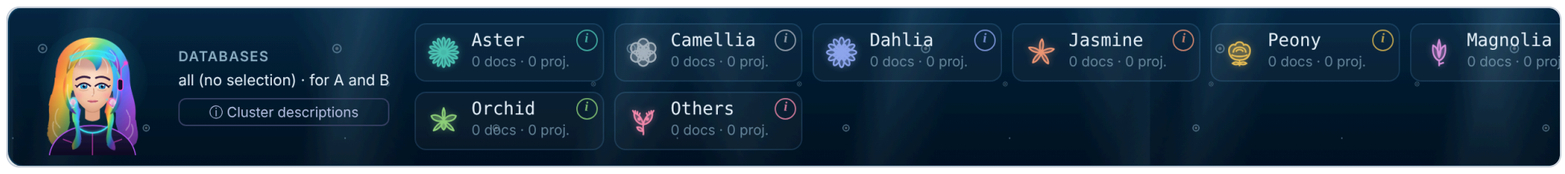
2 Enter your API key

Paste your OpenRouter key (`sk-or-v1-…`) and click **Save**. The browser keeps it. No key yet? How to get one → page 2



3 Choose databases

Click one or more databases. No selection searches all of them. The choice applies to both chats.



ASK A QUESTION pick the chat that fits the question

ⓘ **Where to find them:** the tab **A/B Interaction Interface** shows both chats side by side. The tab **B Interaction Interface** shows only Graph Chat (B), next to the knowledge map, so **Document Chat (A) is only visible on A/B.**

A Document Chat

RAG (retrieval-augmented generation) searches the **text**: ALITA has cut every document into short passages; it finds the passages closest in meaning to your question and writes the answer from them. Best for content, details, quotes and summaries. Narrow the scope under **Advanced settings** (type, method, cluster, years).

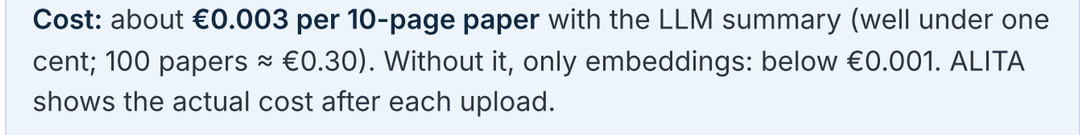


Every claim cites its source as **[1]**, **[2]**. Click a number to see the passage; **Copy** takes the answer with its sources.

> The Mental Model Processes project was written by three authors and submitted to eleven journals [1]. Methodologically, it combines a lab experiment with a multilevel analysis [2].

B Graph Chat

GraphRAG searches the **connections**: on upload, ALITA builds a knowledge graph of people, projects, methods, clusters and document parts, linked by relations such as *authored*, *uses_method* or *cites*. It starts at what your question names and follows these links. Best for relationships: who worked on what, which methods a project uses, who cites whom. The answer shows the path it followed. The graph is based on the SPAR ontologies DoCO, DEO, FaBiO and CiTO plus BIBO for citations between the papers of the collection (page 6).



Under **Advanced settings → Parameters**: filter by entity, relation and period, and set **hops** (1–4) for how far the graph is followed.

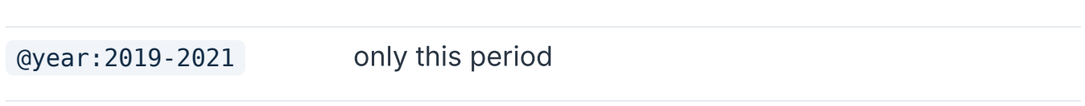
EXAMPLE QUESTIONS

“Who worked with whom on project X?” · more on page 5

EXPLORE AND ADD DATA

Cluster map and 3D view

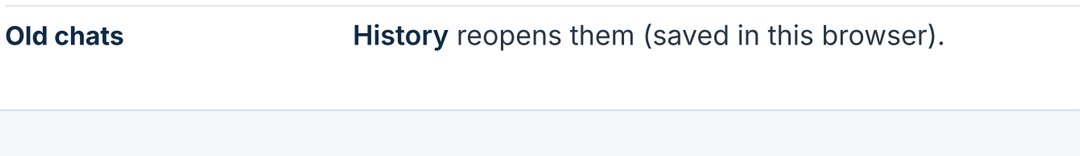
Hover a point to preview, click it to open the document. Click a cluster to limit both chats to it. **Reset view** restores the start view.



3D · time adds the years as depth: `W` `S` forward/back, `A` `D` sideways, drag to look, `Esc` to close, or **Fly through time**.

Add documents

Click **+ Add new data** (or **Documents → + Upload**). Choose the database, the kind of data, optionally a date, then drop the files and click **Upload**. Projects go in as a ZIP of the folder; tick **Generate summary and methods with the LLM** to have them filled in.



Cost: about **€0.003 per 10-page paper** with the LLM summary (well under one cent; 100 papers ≈ €0.30). Without it, only embeddings: below €0.001. ALITA shows the actual cost after each upload.

Uploading uses your OpenRouter key and credits: set it up → page 2

Estimate: 10 pages ≈ 8,000 tokens of text; summary, section labelling and embeddings ≈ 25,000 tokens; GPT-6 Luna at $0.10 / $0.50 per 1M input / output tokens (OpenRouter, Oct 2026).

Prompt filters

`@db:Magnolia`	only this database
`@type:paper`	only papers (or notes, project)
`@year:2019-2021`	only this period
`@cluster:3`	only this cluster

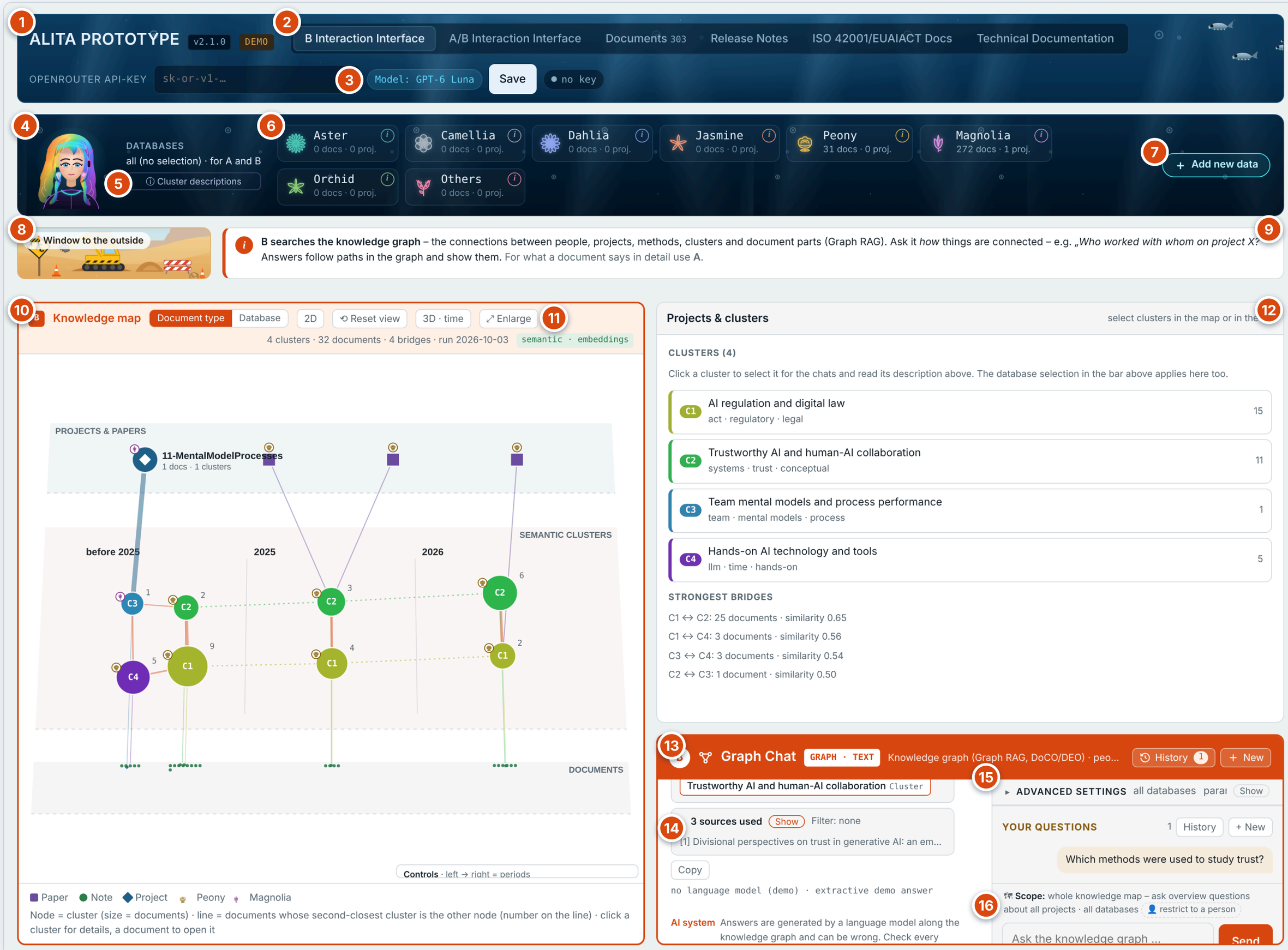
If something looks wrong

No answer	Check the API key is saved (page 2); loosen the filters.
Too narrow	Click **Clear selection** in the database bar.
Off topic	Click **+ New** to start a fresh conversation.
Old chats	**History** reopens them (saved in this browser).

The B Interaction Interface

Knowledge map, projects and Graph Chat on one screen. The numbers match the list below.

B · GRAPH CHAT



TOP BAR AND DATABASES

1 Title and version
Click the version number to open the Release Notes. The DEMO badge appears when the page runs without the backend.

2 Tabs
Switch between **B**, **A/B**, **Documents**, Release Notes and the compliance and technical documentation.

3 API key
Paste your OpenRouter key and click **Save**; the eye shows or hides it. Once saved, the status shows **Key …** with the last four characters. The model is fixed. No key yet? How to get one → page 2

4 ALITA avatar
Follows your cursor. The small **i** buttons on the page say hello and explain the view.

5 Database selection
Shows what is selected; it applies to A and B. **Cluster descriptions** explains all clusters and how the databases share them.

6 Database tiles
Click a tile to select or deselect it; it shows its number of documents and projects. ⓘ opens its contents and cluster descriptions.

7 Add new data
Upload papers, notes or a project ZIP. The selected databases are preset.

8 Window to the outside
Planned external challenge (REQ-20). Opens a concept preview; not active yet.

MAP, PROJECTS AND GRAPH CHAT

9 Info banner
Says what B does: it searches the knowledge graph. For what a document says in detail, use A.

10 Knowledge map
Three layers: projects and papers on top, semantic clusters (C1, C2 …) in the middle, documents below; columns are periods. Circle size = number of documents; lines between clusters = shared documents. Color by **Document type** or **Database**; click a cluster for details, a document to open it.

11 Map controls
2D, **Reset view**, **3D · time** (years as depth), **⇄ Citations** (which papers cite each other) and **Enlarge** to fill the screen.

12 Projects & clusters
Every cluster with its topic, key terms and number of documents; click one to select it for the chats. Below, the strongest bridges between clusters. Project cards and document details open here too.

13 Graph Chat header
History reopens earlier conversations; **+ New** starts a fresh one.

14 Answer
The answer with [n] citations, the **graph path** it followed, the **sources used**, and **Copy**.

15 Advanced settings and your questions
Databases, parameters (entities, relation, period, hops) and the system prompt. Below, your questions; click one to jump to its answer.

16 Scope and question
Shows what the chat searches; **restrict to a person** narrows it. Type your question and click **Send**. Example questions and what works today: use cases → page 5

Use cases for the research proposal

Typical questions, and what the prototype can answer today. ALITA answers from the documents uploaded to it; external literature is outside the prototype.

Current scope of the prototype
This applies to every row below.

- **Within one research project.** Click the project in the map; both chats then answer about that project only.
- **Not yet: between research clusters.** No dedicated comparison of one cluster with another.
- **Not yet: on paper level.** No questions aimed at one single paper or a set of chosen papers.

Status today: **Works** within one project, possibly once data is uploaded · **Partly** parts work; scope, input or rules missing · **Not in prototype** planned for later

USE CASE	TYPICAL QUESTION	WHAT ALITA CAN DO TODAY
YOUR OWN PROJECTS		
1 **Summarise your own knowledge progress across projects**	*"How has my understanding of team mental models developed across my projects since 2015?"*	**Partly – one project at a time** Progress questions are answered from the **knowledge stages** of a project (table of 3, 5 or 10 stages linked to the files), but per project, not across projects. The researchers' own projects are not uploaded yet (to be clarified with Michael).
2 **Identify knowledge leaps and the reasons for them**	*"Where did my thinking change significantly between my early and my recent papers – and why?"*	**Partly – basis to define** Within a project, versions are classified as substantive, rewording, formatting or resubmission. Comparing chosen papers needs paper-level questions (not yet). Open: the basis, e.g. older vs. newer papers of the same authors.
3 **Assess projects and project plans against academic standards**	*"How well does this project plan meet the criteria for a sound research design?"*	**Partly – standards missing** Technically possible. Needed: the project plans (not submitted yet) and the standards, e.g. guidelines as PDF, which can be built into the glossary.
QUESTIONS ABOUT A PROJECT'S DATA AND METHODS		
4 **Explain how the analyses were done**	*"Explain to me how I did the analyses."*	**Works – within one project** Draws on the methods extracted from the files and the analysis documents (scripts, outputs, manuscripts) of the selected project, with sources.
5 **Derive further research questions and loose ends**	*"What further research questions could be derived? Where are the loose ends?"*	**Works – within one project** Answered from the project's limitations, discussion and future-work passages. Not yet across clusters or for one chosen paper.
6 **Find other uses for the data**	*"Can you do anything else with the data?"*	**Works – within one project** Suggestions based on the project's data files, variables and analyses already done. Combining data across projects or clusters is not yet supported.
7 **Uncover research gaps**	*"Which research gaps remain in the 11-MentalModelProcesses project?"*	**Works – tested** Answered from the uploaded documents, with sources. Test results: [to be added].
TOPICS AND IDEAS		
8 **Summarise the state of knowledge on a topic from external sources**	*"What is the current state of research on trust in generative AI?"*	**Not in prototype** External sources were excluded for the initial prototype; answers cover only the uploaded documents. Planned as the **Window to the outside** (REQ-20).
9 **Assess your own research ideas and suggest improvements**	*"Here is my idea: … How strong is it, and how could I improve it?"*	**Works – idea as text** No ideas submitted yet, but none need uploading: paste the idea into the chat. ALITA relates it to the existing documents and cites them.
10 **Support in reaching knowledge leaps**	*"What more should I do with the data? Where do I need to look deeper?"*	**Works – own project needed** Upload the project (**+ Add new data**, ZIP), click it in the map so the chat is limited to it, then ask.

6 work today within one project; some need the researchers' own data or text.

3 work partly: cross-project scope, a comparison basis and academic standards are missing.

1 is outside the prototype: external sources.

Status as of ALITA v2.1.0, October 2026. Example questions are illustrations; answers depend on the uploaded documents.

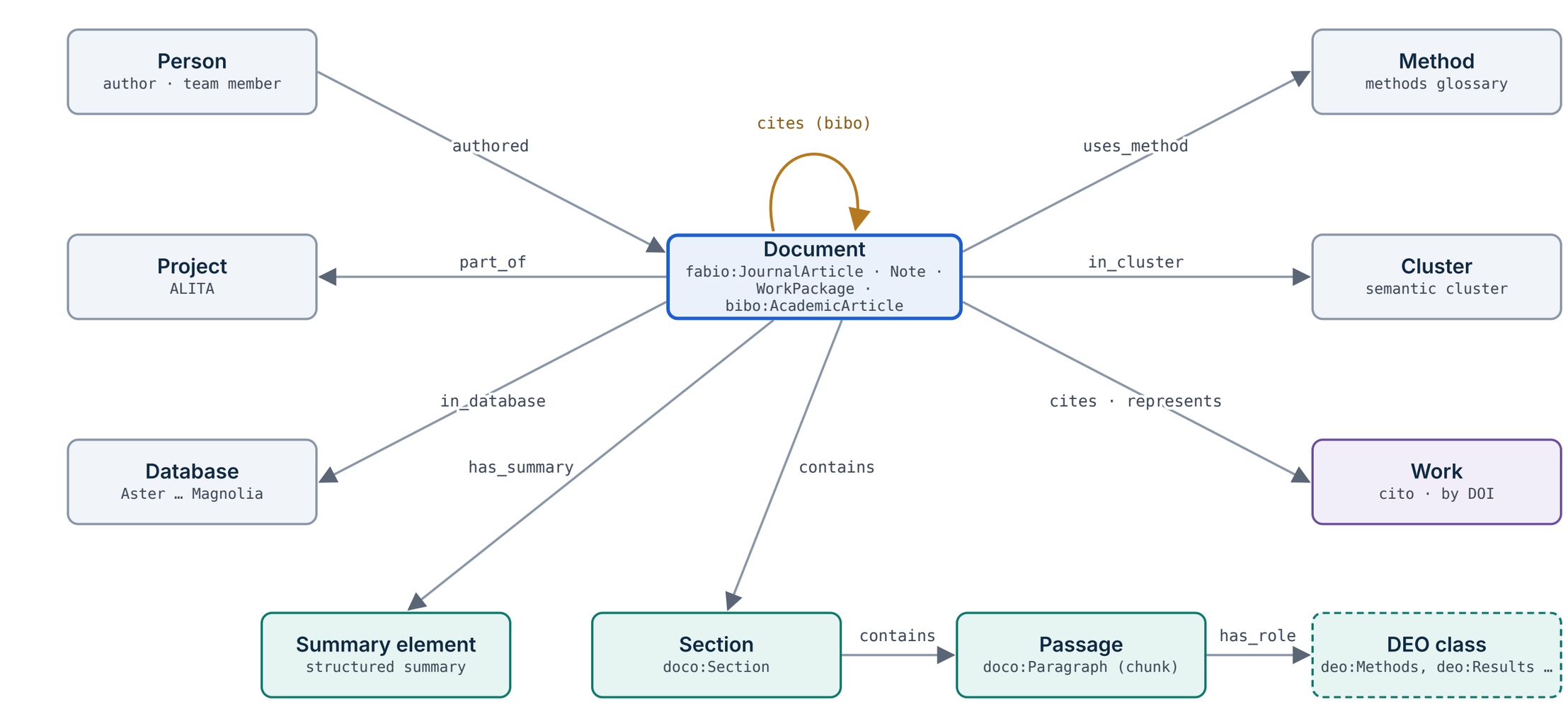
Ontologies behind ALITA

ALITA describes every document with the SPAR ontologies for scholarly publishing and stores the result in a knowledge graph (Neo4j). Graph Chat (B) answers along it; both chats filter by it.

THE ONTOLOGIES

ONTOLOGY	DESCRIBES	HOW ALITA USES IT	EXAMPLES
DEO Discourse Elements Ontology · deo:	The **rhetorical role** of a text part: 31 classes	Every passage, section and summary element gets one or more roles, detected from headings (EN/DE) or by the language model.	Introduction, Methods, Results, Discussion, FutureWork, BibliographicReference
DoCO Document Components Ontology · doco:	The **structure** of a document; imports DEO and the Pattern Ontology	Documents are split into sections and passages in reading order.	Section, Paragraph, Bibliography
Pattern Ontology po:	Pure **containment** structure, the base layer of DoCO	The “contains” links: document → section → passage.	Container, Block
FaBiO FRBR-aligned Bibliographic Ontology · fabio:	The **type** of a publication	Each document is typed by its kind of data.	JournalArticle (paper), Note (notes), WorkPackage (project material)
CiTO Citation Typing Ontology · cito:	**Citations** between works	DOIs in a document's bibliography become “cites” links, so papers in the collection that cite each other are connected.	Document cites Work
ALITA's own project-specific	**Research context** not covered by SPAR	People, projects and teams, databases, the methods glossary and the semantic clusters.	Person, Project, Database, Method, Cluster
BIBO Bibliographic Ontology · bibo:	**Bibliographic description** and citations between documents	Each document gets its BIBO type and ordered author list; the reference lists are matched with the papers of the collection (DOI or title), so papers and projects that cite each other are linked. Shown as **⇄ Citations** in the knowledge map.	AcademicArticle, authorList, cites

HOW IT FITS TOGETHER: THE KNOWLEDGE GRAPH



DoCO / DEO · FaBiO · CiTO · BIBO · ALITA's own — Sections, passages and summary elements each link to their DEO roles (has_role).

Where you see it in the interface

- **Discourse element** filter in both chats: answer only from, e.g., Methods or Results passages. In Graph Chat it is derived from the question if left empty.
- **Graph path** under each Graph Chat answer, e.g. Person → authored → Project → uses_method → Method.
- **Relation** filter (authored, uses_method, cites, contains, in_cluster) and **hops** in Graph Chat.
- **⇄ Citations** in the knowledge map: who cites whom inside the collection; in a document's details, “Cites” and “Cited by”.

Why an ontology

Vector search knows *which document* a passage comes from. The ontology adds *what role it plays* in the argument (method, result, limitation …) and *how documents relate* (authors, methods, citations, clusters). Questions like “Where do the papers discuss limitations?” are then answered along an explicit path instead of by similarity alone.

When each ontology is applied during upload: page 7. Details: tab **Technical Documentation** › Ontology in ALITA. Source: Constantin, Peroni, Pettifer, Shotton, Vitali (2016): The Document Components Ontology (DoCO). Semantic Web 7, 167–181. Ontologies: purl.org/spar/deo · purl.org/spar/doco · purl.org/spar/fabio · purl.org/spar/cito · purl.org/ontology/bibo

How documents get into ALITA

The ingestion pipeline, step by step. The ontology is not applied up front: each passage gets its rhetorical role while it is chunked. Clustering runs on the summaries, not on the passages.

Legend: RULE-BASED no language model · LLM GPT-6 Luna, uses the uploader's OpenRouter key · ONTOLOGY SPAR ontologies, see page 6 · EMBEDDING text → vector

1 Extract RULE-BASED

Text and sections with their headings from PDF, Word, Excel …; metadata: type, date, role, version, duplicates, latest version. Methods are tagged by keyword from the methods glossary.

→ **Document record**
metadata, methods; duplicates and files without text stop here (metadata only)

2 Summarise LLM optional · latest version only

A **structured summary** per work: purpose, design/methodology, findings, limitations, originality, plus methods, keywords and a title.

→ **Structured summary**
5 fields per paper, note or project

3 Chunk and assign roles RULE-BASED DoCO · DEO

Passages of 1,200 characters with 200 overlap; the location in the file is kept. Each passage gets its **DEO role** (Methods, Results, Discussion …) from its section heading, in English or German.

→ **Passages with roles**
e.g. "3 Method" → deo:Methods

4 Label the rest LLM DEO

Passages whose heading gives no role are classified by the language model, at most 60 per document to limit cost.

→ **All passages labelled**
ready to be stored

5 Embed EMBEDDING

Every passage becomes a vector, with a short header (type, project, file). Every paper, note and project also gets a **summary vector** from its title and structured summary.

→ **Qdrant · passages**
vector + section + DEO role

→ **Qdrant · documents & projects**
metadata + summary vector

6 Build the knowledge graph DoCO · DEO · FaBiO · CiTO · BIBO

Document (typed with FaBiO and BIBO) → sections → passages with their DEO roles; people (ordered author list), project, database, methods, the DOIs cited in the bibliography (CiTO) and, after each upload, the citations between documents of the collection, found in the reference lists by DOI or title (BIBO, no LLM).

→ **Neo4j · knowledge graph**
nodes and relations for Graph Chat

7 Cluster K-MEANS LLM for names

After an upload (once at the end of a batch): k-means with cosine similarity on the **summary vectors**, one point per paper, note or project; project files inherit their project's cluster. Names and short descriptions by the LLM, map positions by PCA.

→ **Clusters**
written to Qdrant and the graph

WHO USES WHAT

Document Chat (A) · RAG

Searches the **passage vectors** in Qdrant. The discourse-element filter narrows to, e.g., Methods or Results passages.

Graph Chat (B) · GraphRAG

First finds a **path in the graph** (Neo4j), then searches only the passages of the documents on that path.

Knowledge map

Shows the **clusters** from step 7 and their positions; clicking a cluster limits both chats to it.

Defaults from the backend configuration (ALITA v2.1.0): chunk size 1,200 / overlap 200; summaries for the latest version only; at most 60 LLM-labelled passages per document; clustering automatically after each upload.
Code: backend/app/ingest.py, chunking.py, deo.py, graph.py, clustering.py.